

July 2026 Copper-Conductor Qualification Review

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Deliverable chain: [source_register.xlsx](#) → [technical_findings_matrix.xlsx](#) → [qualification_readout.pptx](#)

Review outcome

The model-derived qualification result contains one release, three conditional results, and one hold. Scores below are read directly from native numeric formula cells in the technical findings matrix; this note does not recalculate them.

LOT A 0 • RELEASE	LOT B 12 • CONDITIONAL	LOT C 8 • CONDITIONAL	LOT D 88 • HOLD	LOT E 44 • CONDITIONAL
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Top three model findings

- 1. LOT-CU-2607-D — 88 / HOLD.** Sulfur and conductivity fail; the trace has a sub-10 mΩ resistance margin and the PCN exception applies.
- 2. LOT-CU-2607-E — 44 / CONDITIONAL.** Lead fails the strictest applicable limit; catalog identity remains unresolved, producing a catalog-quality flag.
- 3. LOT-CU-2607-B — 12 / CONDITIONAL.** Oxygen passes only under MAT-WVR-26-071; identity remains unresolved, so property comparison stays excluded.

Evidence and calculation controls

- The Inputs evidence population remains the same 74 source-register records, with stable IDs and source/detail anchors.
- Only seven measured, protocol-condition-matched property records are selected. HB resistivity of 1.762 μΩ·cm feeds the trace calculation.
- LOT-CU-2607-D calculates to 155.344 mΩ at 85 °C, leaving a 4.656 mΩ margin to the 160 mΩ limit.
- Chemistry is separate and uses the strictest applicable published limit; MAT-WVR-26-071 changes only LOT-CU-2607-B oxygen acceptance.
- Risk weights, thresholds, service factors, and controlled limits are named native numeric Inputs cells.

Sensitivity interpretation

Scenario	LOT D	LOT B	Decision signal
Current controlled basis	88 • HOLD	12 • CONDITIONAL	Current controlled basis
No oxygen waiver	88 • HOLD	44 • CONDITIONAL	B oxygen fails without waiver
No process change notice	80 • HOLD	12 • CONDITIONAL	D thickness fails without PCN
95 C service stress	88 • HOLD	12 • CONDITIONAL	D resistance fails at 95 °C

Action register

Action	Owner	Closure criterion	Evidence anchor
Contain LOT-CU-2607-D and resolve sulfur/conductivity failures	Nora Patel / Meredith Chen	Validated disposition plus acceptable repeat evidence or approved rejection	ASSAY-0726-04; LR-COND-0726-05
Resolve LOT-CU-2607-E lead failure and η -Cu ₃ Sn identity gap	Evan Brooks	Supplier phase/composition evidence and lead disposition accepted	ASSAY-0726-05; MAT-CU ₃ SN-ETA
Close LOT-CU-2607-B waiver and identity trace	Nora Patel / Evan Brooks	Waiver remains registered and supplier identity corroboration is attached	ASSAY-0726-02; MAT-WVR-26-071
Define percent-IACS protocol conditions before next refresh	Dr. Lena Ortiz	Controlled protocol revision specifies conductivity test conditions	PROTO-CU-77

Verification

Model checks: 18/18 PASS. **Scenario row checks:** 4/4 PASS. **Visible formula errors:** none detected.

Primary review artifacts: [Source register](#) • [Technical findings matrix](#) • [Qualification readout](#)